THE PICARD GROUP OF THE CONTRACTION OF AN EXCEPTIONAL CURVE OF THE FIRST KIND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the restriction map with the degree and the reduction to the statement on Picard groups, the verification that the pullback is an isomorphism onto the invertible sheaves restricting trivially to the curve, and the verification that the pushforward inverts the pullback after reduction to a regular local base, together with the explicit splitting.

Editorial note. The label lemma-pic-blowup names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-pic-blowup –start 4776 –end 4836 –local-id 28086 returned RESULT=OK (L28086 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The Picard group of the projective line and the degree of the exceptional divisor. Divisors, the lemma computing the Picard group of a projective space, together with the defining property of an exceptional curve of the first kind that the restriction of its associated invertible sheaf has degree minus one. Supplies the identification of the last map with restriction to the curve and the computation of the degree of the restriction of the associated invertible sheaf.
- Cohomology of the structure sheaf of the blowup and triviality of invertible sheaves over a regular local base. Same chapter, the lemma computing the cohomology of the structure sheaf of the blowup and, in its last part, the statement that the pushforward of the structure sheaf is the structure sheaf, together with the lemma describing the Picard group of the blowup of a regular local ring. Supplies the isomorphism between a module on the contraction and the pushforward of its pullback, and the triviality of invertible sheaves after restricting to the local ring of the contraction at the image point.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Flat base change for pushforward, pullback and the adjunction map. Cohomology of Schemes, the flat base change lemma, applied to the computation of the pushforward and of the adjunction map. Supplies the reduction of the two claims to the local ring of the contraction at the image point of the curve.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 4821; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BXJ: sources/stacks-chow/tags.txt, source lines 14982–14982. Prerequisite pointer invoked at proof line(s) 4793; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C5G: sources/stacks-resolve/pinned-original-source.tex, source lines 756–756. Prerequisite pointer invoked at proof line(s) 4825; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AGS: sources/stacks-resolve/pinned-original-source.tex, source lines 786–786. Prerequisite pointer invoked at proof line(s) 4809; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $b : X \to X'$ be the contraction of an exceptional curve of the first kind $E \subset X$. Then there is a short exact sequence*

$$0 \to \mathrm{Pic}(X') \to \mathrm{Pic}(X) \to \mathbf{Z} \to 0$$

where the first map is pullback by b and the second map sends $\mathcal{L}$ to the degree of $\mathcal{L}$ on the exceptional curve E. The sequence is split by the map $n \mapsto \mathcal{O}_X(-nE)$.

Proof. Since $E = \mathbf{P}^1_k$ we see that the Picard group of E is $\mathbf{Z}$, see Divisors, Lemma 0BXJ. Hence we can think of the last map as $\mathcal{L} \mapsto \mathcal{L}|_E$. The degree of the restriction of $\mathcal{O}_X(E)$ to E is -1 by definition of exceptional curves of the first kind. Combining these remarks we see that it suffices to show that $\mathrm{Pic}(X') \to \mathrm{Pic}(X)$ is injective with image the invertible sheaves restricting to $\mathcal{O}_E$ on E.

Given an invertible $\mathcal{O}_{X'}$-module $\mathcal{L}'$ we claim the map $\mathcal{L}' \to b_*b^*\mathcal{L}'$ is an isomorphism. This is clear everywhere except possibly at the image point $x \in X'$ of E. To check it is an isomorphism on stalks at x we may replace X' by an open neighbourhood at x and assume $\mathcal{L}'$ is $\mathcal{O}_{X'}$. Then we have to show that the map $\mathcal{O}_{X'} \to b_*\mathcal{O}_X$ is an isomorphism. This follows from Lemma 0AGS part (4).

Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module with $\mathcal{L}|_E = \mathcal{O}_E$. Then we claim (1) $b_*\mathcal{L}$ is invertible and (2) $b^*b_*\mathcal{L} \to \mathcal{L}$ is an isomorphism. Statements (1) and (2) are clear over $X' \setminus \{x\}$. Thus it suffices to prove (1) and (2) after base change to $\mathrm{Spec}(\mathcal{O}_{X',x})$. Computing b_* commutes with flat base change (Cohomology of Schemes, Lemma 02KH) and similarly for b^* and formation of the adjunction map. But if X' is the spectrum of a regular local ring then $\mathcal{L}$ is trivial by the description of the Picard group in Lemma 0C5G. Thus the claim is proved.

Combining the claims proved in the previous two paragraphs we see that the map $\mathcal{L} \mapsto b_*\mathcal{L}$ is an inverse to the map

$$\mathrm{Pic}(X') \longrightarrow \mathrm{Ker}(\mathrm{Pic}(X) \to \mathrm{Pic}(E))$$

and the lemma is proved. □